

EXAM POV NOTES (O&A FORMAT)**Q1: What constitutes an AI system, and how do agents and environments interact within it? [10 Marks]**

- An AI system is fundamentally composed of an agent and its environment, forming the core of any AI concept.
- These systems are built on principles that enable them to perceive their world and actively operate within it.
- An agent is defined as anything that can sense its surroundings using sensors and subsequently performs actions through effectors or actuators.
- The environment encompasses everything that is external to the agent, serving as the world or situation where the agent lives and operates.
- The environment provides the agent with stimuli to sense and the context upon which it can act.
- This interaction is a continuous perceive-then-act loop, where the agent gathers information and then chooses a response.
- Sensors (like eyes, ears, cameras) allow the agent to gather input, while actuators (like hands, legs, motors) enable it to perform physical or digital actions.
- The agent's goal is to make rational decisions, choosing actions most likely to move it closer to its objectives.
- The environment is not always solitary; it may contain other agents, which can be of the same or different types, adding complexity to interactions.
- Understanding this fundamental composition and interaction is crucial for designing any AI system, from simple to complex.

Q2: Describe the evolution of intelligent agents from simple reflexes to sophisticated decision-makers, outlining their distinct characteristics. [10 Marks]

- Simple Reflex Agents are the most basic, reacting solely based on the current percept and adhering to "if-then" rules.
- They possess no memory of past events, meaning their decisions are made in the immediate present.
- Rationality for these agents is only possible in completely observable environments, where the current state provides all necessary information.
- Model-Based Reflex Agents introduce an internal "model" or representation of the world to inform their actions.
- They maintain an internal state, which represents unobserved aspects of the current situation derived from percept history.
- This allows them to operate effectively in partially observable environments by remembering past states and predicting changes.
- Goal-Based Agents advance further by having specific goals they strive to achieve, allowing for more flexible planning.
- They choose actions by thinking ahead about future outcomes and selecting paths that lead them towards their desired situations.
- Utility-Based Agents are the most advanced, possessing preferences or "utility" for various states, enabling complex decision-making.
- They can weigh conflicting goals and uncertainties, choosing actions that maximise overall utility or "happiness" for the best possible outcome.

Q3: Explain the PEAS framework used by AI designers to specify an agent's task. [10 Marks]

- The PEAS framework is a simple yet powerful four-part tool used by designers to perfectly specify an agent's task, serving as a comprehensive design checklist.
- P stands for Performance Measure, which defines the criteria by which the agent's success is determined and evaluated.
- For a self-driving car, this measure could include metrics like safety, speed, and adherence to traffic laws.
- E stands for Environment, which refers to the specific world or context in which the agent operates.
- For example, the environment for a self-driving car includes roads, traffic conditions, pedestrians, and weather.
- A stands for Actuators, which are the physical or logical components enabling the agent to perform actions.
- A self-driving car's actuators include steering, acceleration, braking, and signaling mechanisms.
- S stands for Sensors, representing how the agent perceives or gathers information about its environment.
- For a self-driving car, sensors might include cameras, radar, lidar, GPS, and odometers.
- By clearly defining these four elements, designers precisely set the stage for the agent, articulating its purpose and operational context.
- This framework ensures clarity in what the agent needs to achieve and how it will interact with its surroundings.

Q4: Discuss the various properties used to classify and define an agent's environment. [10 Marks]

- Discrete / Continuous: An environment is discrete if it has a limited, distinct set of states (e.g., chess), while continuous environments have a fluid, unbounded range of states (e.g., driving).
- Observable / Partially Observable: An observable environment allows the agent to know its complete state from percepts at any time (e.g., chessboard), whereas partially observable means some parts are hidden (e.g., what's around a corner).
- Static / Dynamic: A static environment remains unchanged while the agent acts, simplifying decisions, but a dynamic environment changes independently, requiring continuous monitoring (e.g., moving traffic).
- Single Agent / Multiple Agents: This property describes whether the environment contains only the agent or also other agents, which can be of the same or different kinds, influencing strategy and cooperation.
- Accessible / Inaccessible: An environment is accessible if the agent's sensory apparatus can fully perceive its complete state, providing comprehensive information.
- Deterministic / Non-deterministic: A deterministic environment's next state is entirely predictable from the current state and agent actions, while a non-deterministic one has unpredictable outcomes due to external factors or inherent randomness.
- Episodic / Non-episodic: In an episodic environment, each perceive-act cycle is independent, with no impact on future episodes, making tasks simpler (e.g., spam filtering).
- A non-episodic environment means actions have long-term consequences, requiring the agent to plan ahead and consider the impact of its choices over time.

- Each of these properties fundamentally changes the problem an agent needs to solve, impacting its design and complexity.
- Classifying an environment by these properties helps AI designers understand the challenges and requirements for building effective agents.

Q5: What is rationality in AI, and what factors determine an agent's rationality? [5 Marks]

- Rationality in AI signifies the status of being reasonable, sensible, and possessing a good sense of judgment.
- It is fundamentally about an agent making the best possible decision to achieve the best possible outcome.
- A rational agent considers what it knows and perceives, then selects the action most likely to bring it closer to its defined goal.
- An ideal rational agent maximises its performance measure by performing expected actions, based on its percept sequence and built-in knowledge.
- An agent's rationality is determined by its performance measures (degree of success), its percept sequence (what it has perceived so far), and its prior knowledge about the environment.
- Furthermore, the actions the agent can carry out also factor into its rational behaviour, influencing the choices it can make.

Q6: Explain the key terminology associated with AI agents. [5 Marks]

- Performance Measure: This is the specific criterion that quantifies and determines how successful an agent is in achieving its objectives.
- Behavior of Agent: This refers to the particular action an agent performs in response to a given sequence of percepts it has received.
- Percept: A percept is the individual, instantaneous sensory input that an agent receives from its environment at any specific moment.
- Percept Sequence: This constitutes the complete historical record of all sensory inputs, or percepts, that an agent has ever perceived up to the current time.
- Agent Function: This is a conceptual map that mathematically or logically defines how an agent's entire percept sequence translates into a specific action.
- Rationality: This refers to the agent's ability to make decisions that lead to the best possible outcome, based on its perceptions and knowledge.

Q7: Describe the basic structure of intelligent agents. [5 Marks]

- The fundamental structure of an intelligent agent can be conceptually broken down into two main components.
- It is viewed as the combination of an Agent = Architecture + Agent Program.
- Architecture refers to the underlying machinery or hardware platform on which the agent executes its operations.
- This machinery provides the computational and physical means for the agent to function.
- The Agent Program is an implementation of the agent function, essentially the software logic that defines how the agent maps percepts to actions.
- This program dictates the agent's intelligent behaviour, enabling it to process information and make decisions.

Q8: What are the different categories of agents based on their physical embodiment? [5 Marks]

- Human Agents: These agents use sensory organs like eyes, ears, nose, and skin as their primary sensors.
- Their effectors, or actuators, include biological organs such as hands, legs, and the mouth, enabling them to act on the environment.
- Robotic Agents: These agents replace biological senses with artificial ones, employing cameras and infrared range finders for perception.
- Their actions are carried out using various motors and mechanical actuators.
- Software Agents (Softbots): These agents operate in artificial or simulated environments, receiving input as encoded bit strings or data.
- Their "actions" are typically executed as programs or commands within a digital system, for example, scanning online preferences.
- Despite their diverse forms, all categories adhere to the core principle of sensing their environment and subsequently acting upon it.
- Some softbots can also operate in mixed real and artificial environments.

Q9: What is an Agent? [2 Marks]

- An agent is fundamentally defined as anything capable of perceiving its environment through sensors.
- Following perception, it acts upon that environment through effectors or actuators.

Q10: What is an Environment in the context of AI? [2 Marks]

- The environment is everything in the world that surrounds the agent, distinct from the agent itself.
- It serves as the situation where the agent exists and operates, providing stimuli for perception and opportunities for action.

Q11: Define a Condition-Action Rule. [2 Marks]

- A Condition-Action Rule is a simple rule that directly maps a specific perceived state or condition to a predefined action.
- It dictates an "if this, then that" response, characteristic of simple reflex agents.

Q12: What is an Internal State in Model-Based Reflex Agents? [2 Marks]

- An Internal State is a representation within a model-based agent of the unobserved aspects of its current environment.
- This state is updated and maintained based on the agent's percept history and its knowledge of how the world changes and how its actions impact it